

Signals & Systems (EE504)
Pre-Reading Material – Lecture 18

Digital Filters

Estimated Reading Time: 20–25 minutes

1. Why Do We Study Digital Filters?

Digital filters are used to **process discrete-time signals** by modifying or removing unwanted components.

They are widely used in:

- Audio and speech processing
- Image and video processing
- Communication systems
- Biomedical signal processing
- Noise reduction
- Sensor and measurement systems

Before studying digital filters, students should be familiar with **discrete-time signals, convolution, and basic system concepts**.

2. What is a Digital Filter?

A **digital filter** is a discrete-time system that processes an input sequence to produce a desired output sequence.

The main purpose is to:

- Remove unwanted noise
- Pass desired frequency components
- Suppress unwanted frequencies
- Modify the characteristics of a signal

Example: A noisy speech signal can be processed using a digital filter to reduce unwanted high-frequency noise.



3. Basic Idea of Filtering

A signal may contain different frequency components.

A filter can be designed to:

- **Allow** certain frequency components to pass
- **Attenuate** or reject other frequency components

This leads to different types of frequency-selective filters.

4. Types of Digital Filters

Based on their frequency-selective behavior, common filters are:

Filter	Main Function
Low-Pass Filter (LPF)	Allows low frequencies and attenuates high frequencies
High-Pass Filter (HPF)	Allows high frequencies and attenuates low frequencies
Band-Pass Filter (BPF)	Allows a selected range of frequencies
Band-Stop Filter (BSF)	Rejects a selected range of frequencies

Think: If a signal contains useful low-frequency information and unwanted high-frequency noise, which filter would be appropriate?

5. FIR and IIR Filters

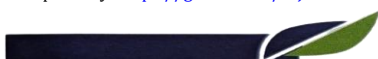
Digital filters are commonly classified into two major categories:

FIR — Finite Impulse Response

- Has a finite-duration impulse response.
- Generally uses present and past input samples.
- Does not require feedback.
- Can provide linear-phase characteristics under suitable design conditions.

IIR — Infinite Impulse Response

- Has an impulse response that theoretically continues indefinitely.
- Uses present/past input samples and feedback from previous output samples.
- Can achieve desired filtering characteristics with fewer coefficients in many applications.



6. Filter Structure

A digital filter can contain:

- **Delay elements** – store previous samples
- **Multipliers** – apply filter coefficients
- **Adders** – combine signal components
- **Feedback paths** – commonly present in IIR filters

Understanding these building blocks helps in understanding the implementation of FIR and IIR filters.

7. Important Terms to Know

Before the lecture, become familiar with:

- **Passband:** Frequency range that is allowed to pass with relatively little attenuation.
- **Stopband:** Frequency range that is significantly attenuated.
- **Transition band:** Region between the passband and stopband.
- **Cutoff frequency:** Frequency used to indicate the boundary of a filter's transition from passing to attenuating.
- **Frequency response:** Describes how a system responds to different frequency components.

8. Applications

Digital filters are used in practical systems such as:

- Noise cancellation
- Speech enhancement
- ECG and biomedical signal processing
- Audio equalizers
- Wireless communication
- Radar and sonar
- Image processing
- Seismic signal processing

9. Learning Outcomes

After studying this topic, you should be able to:

1. Explain the purpose of digital filtering.
2. Differentiate between LPF, HPF, BPF, and BSF.
3. Distinguish between FIR and IIR filters.
4. Identify the basic components of a digital filter.
5. Explain important filter characteristics such as passband, stopband, and transition band.

Think Before the Lecture

1. Why is filtering necessary when a signal already contains useful information?
2. What type of filter would you use to remove high-frequency noise from a signal?
3. What is the basic difference between FIR and IIR filters?
4. Why are delay elements required in digital filters?
5. Can a filter completely remove an unwanted frequency component in a practical system?

Quick Recall

Signal → Digital Filter → Modified/Filtered Signal

The key idea is: **A digital filter selectively modifies the frequency components of a discrete-time signal.**

